

Jammi Prasanthi Sirasani

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EDUCATION

• Saint Louis University, Saint Louis, MO

Fall 2022-May 2024

M.Sc. (Bioinformatics & Computational Biology) 3.96/4.0

- **Courses:** Object Oriented Programming (Python), Introduction to Bioinformatics-I/II, Genomics, Advanced Microbiology, Analytics and Statistical Programming (R language), Data Structures (with C++), Molecular techniques, Algorithms for Bioinformatics, Introduction to Machine Learning, Current topics in Biology.

Study in India

• Acharya N. G. Ranga Agricultural University , India

Fall 2018- 22

Ph.D. (Entomology) 9.25/10

- Secured the highest marks in the Ph.D. Entrance exam of Acharya N G Ranga University.
- Merit Scholarship from BAYER CROP SCIENCES for three years of Ph.D.
- Studied insect-species diversity based on classical taxonomy and molecular markers like mitochondrial cytochrome oxidase subunit I, histone protein (H4), etc. using various bioinformatics tools.
- Evaluated the effect of insecticides and meteorological factors against mango leafhoppers.

• Birsa Agricultural University, India

Fall 2014-16

M.Sc. (Entomology) 9.10/10

- An academic and a Chancellor's gold medal for securing the highest OGPA
- Qualified National Eligibility Test for university-level teaching conducted by the Indian Council of Agricultural Research
- Conducted initial and advanced field trials to screen the genetic tolerance/resistance of various germplasms of chickpea against pod borers as a part of the All India Coordinated project.

• Acharya N. G. Ranga Agricultural University

Fall 2010-14

B.Sc. (Agriculture) 8.38/10

WORK EXPERIENCE

I. Jr. Bioinformatics Scientist March 2026 – (Full-time Regular)

Location: Psomagen Inc. (CLIA-certified and CAP-accredited), Rockville, Maryland.

- QC of the sequenced data from the sequencing team.
- Performing Bioinformatics analysis on short/long reads (WGS, WES, bulk RNA, single-cell RNA, Spatial transcriptomics) based on the client's requirement.

II. Research Technician II July 8th, 2024 – March, 2026 (Full-time Regular)

Location: Dr. Kiran Mahajan's Lab, Department of Surgery-Urology, Washington University, St. Louis

Exploring the role of HOXB13 in Prostate Cancer

- Planning and conducting experiments to explore the Tumor Microenvironment (TME) of prostate cancer in the context of HOXB13 and its mutations.

- **Skills**
 - **Bench work:** mice handling and genotyping, preparing samples for high-throughput sequencing (bulkRNA and scRNA-seq, ChIP-seq), Tissue Micro Array and flow cytometry, RNA extraction, cDNA synthesis, qRT-PCR, cell cultures, and ImmunoHistoChemistry (IHC), developing new drugs and testing them towards immunotherapy in prostate cancer.
 - **Bioinformatics:** Analyzing the data from flow cytometry (FlowJo), bulk RNA-seq, scRNA-seq (Seurat), ChIP-seq, Tissue Micro Array (TMA) by QUPATH.
 - **Related Publications:** **research article under consideration in Nature Communications and another research article sent for initial review.**

III. Volunteer & Graduate Student Assistant

Volunteer: August 23, 2023, to December 15, 2023 (4 months)

Graduate Student Assistant: February 02, 2024, to May 30, 2024 (4 months)

Location: Dr. Tae-Hyuk Ahn's Lab, Department of Computer Science, Saint Louis University, St. Louis

Responsibilities:

- **Project name: Blood and tissue microbiome analysis:** I have compiled literature concerning bioinformatics protocols from quality control to downstream analysis (associated controversies and challenges) and ML models for Microbiome analysis of blood and tissue to diagnose cancer.
- **Related publication**
Sirasani, J.P*., Gardner, C., Jung, G., Lee, H. and Ahn, T.H., 2025. Bioinformatic approaches to blood and tissue microbiome analyses: challenges and perspectives. *Briefings in bioinformatics*, 26(2), p.bbafl176.

IV. Bioinformatics Assistant Internship

Duration: May 22, 2023, to December 15, 2023

Full-time: May 22, 2023, to August 22, 2023 (3 months)

Part-time (20 hrs/week): August 23, 2023, to December 15, 2023 (4 months)

Location: Dr. Dilip Shah's Lab, Donald Danforth Plant Science Center, Saint Louis, MO, in collaboration with Covercress Inc., Saint Louis, MO

Responsibilities:

- Experimental testing of plant elicitor peptides for antifungal efficacy, involving fungal culture, cultivation of Pennycress under controlled conditions, and post-inoculation application of peptides.
- Conducted bulk RNA-seq data analysis on two Pennycress varieties infected with fungi at two time points.
- Utilized bioinformatics tools such as psiBLAST, Hmmer, blast, and Jackhmmer to identify homologous elicitor peptides in Brassica napus, enriching the understanding of peptide roles across species.
- Addressing the challenge of unrecognized Pennycress gene IDs by tools like g:Profiler and DAVID, I successfully retrieved genomic coordinates for elicitor peptide sequences using STAR and BLASTn with mRNA data, and developed a custom annotation file to enable accurate gene

expression quantification for these and other crucial, unannotated genes in the Pennycress reference genome.

○ **Related Publication**

Codjoe, J.M., Kujur, A., **Sirasani, J.P.***, Shamin, A., Sauer, T., Rai, K., Ulmasov, T., Chopra, R. and Shah, D.M., 2025. Fungal susceptibility and early flowering in pennycress (*Thlaspi arvense*) are conferred by naturally occurring mutations in histone demethylase Jumonji 14. *The Plant journal: for cell and molecular biology*, 125(1), p.e70635.

V. **Teaching Associate and later promoted to Assistant Professor**

Duration: October 2016 to July 2018

Responsibilities:

- Taught Entomology courses (Fundamentals of Entomology, Introduction to Biology, Introduction to Nematology, and Principles of Integrated Pest Management) to undergraduate students enrolled in B.Sc.(Agriculture)

SKILLS

- Programming languages: - Python, C++, R Programming, SQL, RShiny.
- Microsoft office (excel, power point, word), Linux, Docker, GraphPad Prism, Flowjo, Qupath, Version control with Git, Jupyter Notebook, Nextflow.
- Data mining with Blast, psiblast, HMMer, JackHMMer, e-utilities.
- Genome assembly of short reads (using velvet and spades) and long reads (using canu).
- Analyze NGS data:- Metagenomics, Transcriptomics (bulk mRNA, single-cell RNA-seq and micro RNA seq analysis), Variant calling (Germline and Somatic variations, GATK), ChIP-seq, etc.
- ANOVA, Correlation, Regression, Experimental designs, Differential gene expression, etc.
- Machine learning: Scikit-learn, Keras, Tensorflow.

References Contacts

1. **Dr. Tae-Hyuk (Ted) Ahn,**

Associate Professor
Department of Computer Science,
Room 234 B,
Interdisciplinary Science and Engineering Building,
Saint Louis University, St. Louis, MO,
Email: ted.ahn@slu.edu
Ph: 314-977-3633
(I worked in the lab of Dr. Ted from August-May 2024 in the project “Blood and tissue microbiome”)

2. **Dr. Dilip Shah,**

Principal Investigator,
Donald Danforth Plant Science Center,
St. Louis, MO,
Email: dshah@danforthcenter.org
Ph: 314-587-1482
(I worked as a Bioinformatics assistant in the lab of Dr. Shah between May 2022 and December, 2022 in a project involving Bulk RNA-seq data analysis)

3. **Dr. Jennette S. Codjoe,**

Plant Biology Lead,
CoverCress Inc.,
St. Louis, MO
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Linkedin: [linkedin.com/in/jenny-shoots-codjoe-phd-39542191](https://www.linkedin.com/in/jenny-shoots-codjoe-phd-39542191)

(Dr. Jenny was my supervisor with whom I have closely worked at Dr. Shah's lab)